E-Classroom	**Name:**					
	Gr 10		**Date:**		**Time:**	$1\frac{1}{2}$ **HR**
CAPS Reference	**Function**					
Topic	**Parabolic Function**					

1. GOT IT! [15 MIN]

Basic standard:

$$f(x) = ax^2 + q$$

$f(x)$: – represents the $y-$ value

a: – indicates if the function is positive or negative

x^2: – represents the 2 $x-intercepts$

q: – represents the $y-intercept$

– indicates whether the function will move up or down

– indicates the turning point of the function

– indicates whether the function has maximum or minimum value

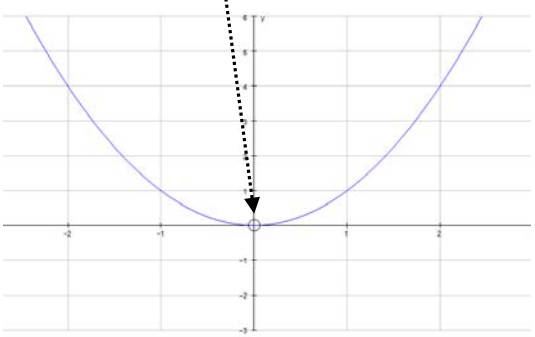
Parabolic function.

Positive function; $a > 0$

Domain: $x \in (-\infty; \infty)$

Range: $y \in (0; \infty)$

Function has a **minimum value**

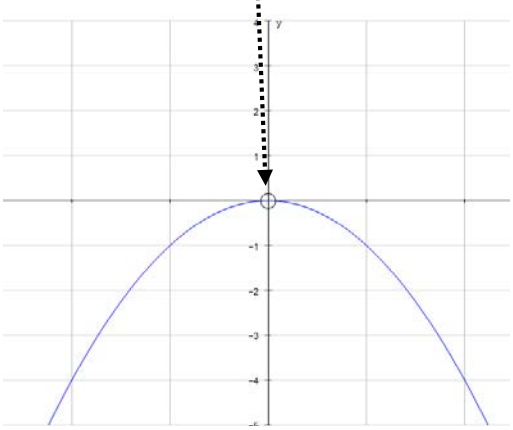


Negative function; $a < 0$

Domain: $x \in (-\infty; \infty)$

Range: $y \in (-\infty; 0)$

Function has a **maximum value**



Drawing of a Parabolic function.

$$f(x) = x^2 - 4$$

$\underline{y - intercept; x = 0}$

$y = (0)^2 - 4$

$= -4$

$(0; -4)$

$\underline{x - intercept; y = 0}$

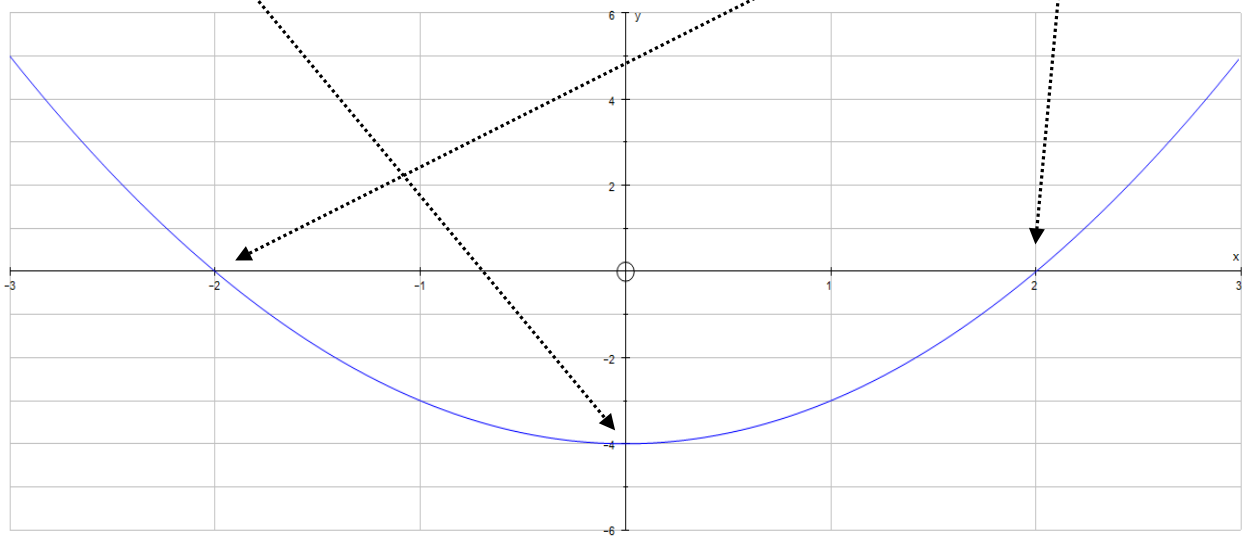
$0 = x^2 - 4$

$0 = (x + 2)(x - 2)$

$x + 2 = 0 \quad or \quad x - 2 = 0$

$x = -2 \quad or \quad x = 2$

$(-2; 0) \quad or \quad (2; 0)$



Important information about the function.

$a > 0$

This indicates that the function is positive.

It will have a minimum value

Turning point $(0; -4)$

The turning point indicates when the function changes direction.

$f(x) > 0$ when $x > 2$ *or* $x < -2$

$f(x) < 0$ when $-2 < x < 2$

Domain:

$x \in (-\infty; \infty)$

Range:

$y \in (-4; \infty)$

$q = -4$

This indicates that the function has moved 4 units down from the original function $y = x^2$

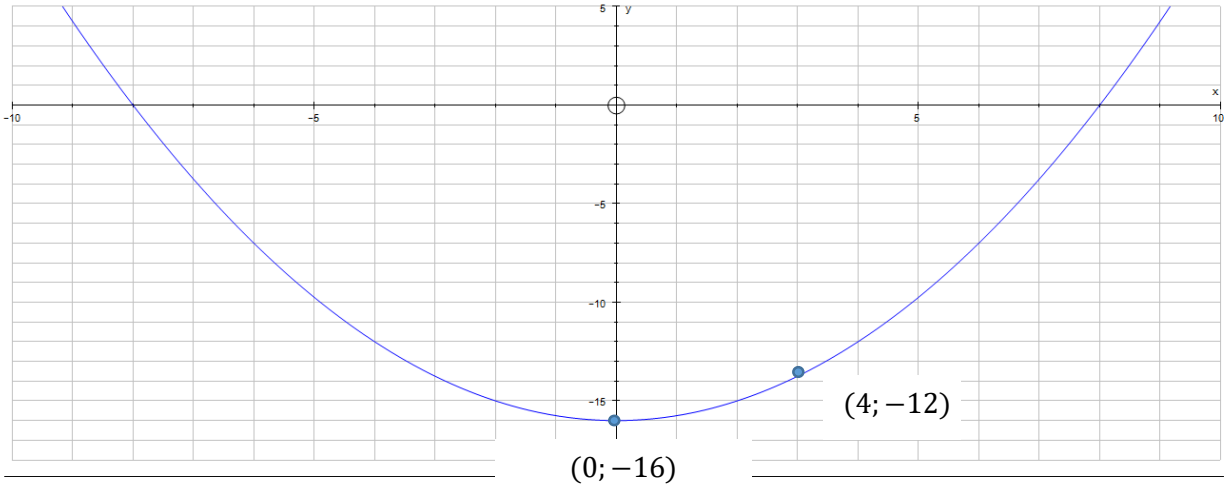
Symmetry line $y = 0$

Indicates where the function is divided in half.

Determine the equation of a Parabolic function

$$f(x) = ax^2 + q$$

Given: The turning point and a coordinate.



Step 1: Substitute the turning point $(0; -16)$

$y = ax^2 + q$

$y = ax^2 - 16$

Step 2: Substitute the given coordinates $(4; -12)$

$y = ax^2 - 16$

$-12 = a(4)^2 - 16$

$-12 = 16a - 16$

$-12 + 16 = 16a$

$4 = 16a$

$\frac{4}{16} = a$

$\frac{1}{4} = a$

Step 3: Substitute a in to the equation.

$$f(x) = \frac{1}{4}x^2 - 16$$

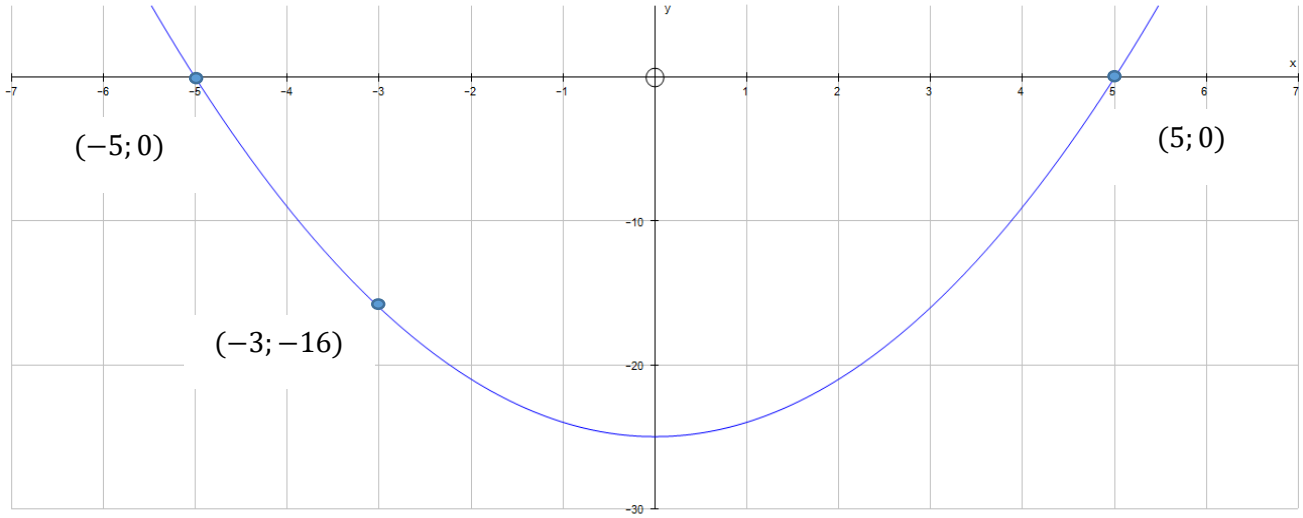
Determine the equation of a Parabolic function

$$f(x) = ax^2 + q$$

Given: $x - intercepts$ and a coordinate.

Then we use the following formula

$$f(x) = a(x - x_1)(x - x_2)$$



Step 1: Substitute the $x - intercepts(-5; 0)\ and\ (5; 0)$

$y = a(x - x_1)(x - x_2)$

$y = a(x - (-5))(x - 5)$

$y = a(x + 5)(x - 5)$

Step 2: Substitute the given coordinates $(-3; -16)$

$-16 = a(-3 + 5)(-3 - 5)$

$-16 = a(2)(-8)$

$-16 = -16a$

$\frac{-16}{-16} = a$

$1 = a$

Step 3: Substitute a in to the equation.

$y = a(x + 5)(x - 5)$

$y = 1(x + 5)(x - 5)$

$y = 1(x^2 - 25)$

$y = x^2 - 25$

Equation

$$f(x) = x^2 - 25$$

2. GO AHEAD [60 MIN]

2. Sketch the following on a set of axis and answer the questions that follow.

2.1 $f(x) = -x^2 + 4$

a) Write down the range of f

b) Write down the turning point of f

c) Determine the value of x for which $f(x) > 0$

d) Write down the equation of h, where $h(x) = f(x) + 2$

2.2 $g(x) = -3x^2 + 27$

a) Write down the range of f

b) Write down the turning point of g

c) Determine the value of x for which $g(x) \leq 0$

d) Write down the equation of the symmetry line.

e) Write down the equation of h, where $h(x) = -g(x) - 2$

2.3 $f(x) = \frac{1}{2}x^2 - 1$

a) Write down the range of f

b) Write down the turning point of f

c) Determine the value of x for which $f(x) \geq 0$

d) Write down the range of h, where $h(x) = f(x) + 3$

2.4 $g(x) = -\frac{1}{3}x^2 + 2$

a) Write down the range of g

b) Write down the turning point of g

c) Determine the value of x for which $g(x) > 0$

d) Write down the equation of the symmetry line.

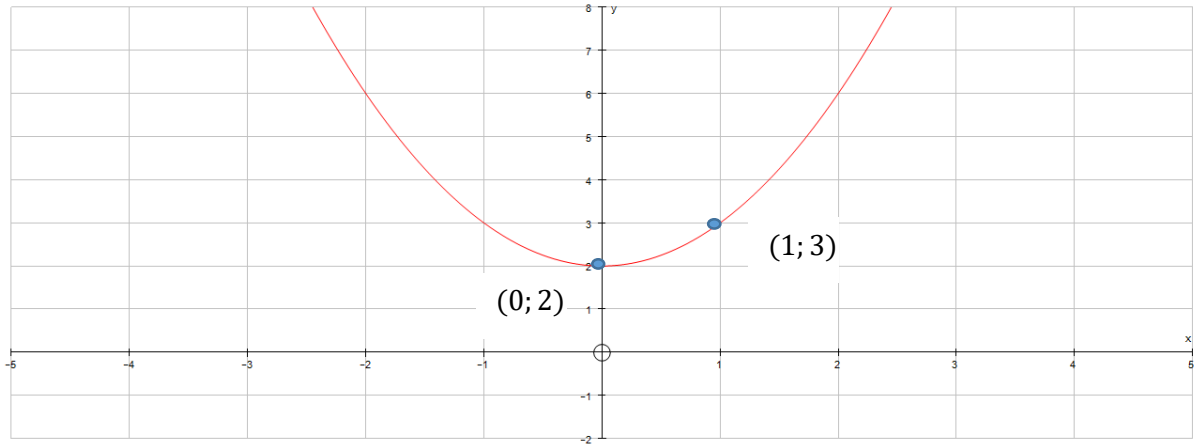
e) Write down the range of h, where $h(x) = -g(x) + 1$

3. CHECK YOUR WORK![15 MIN]

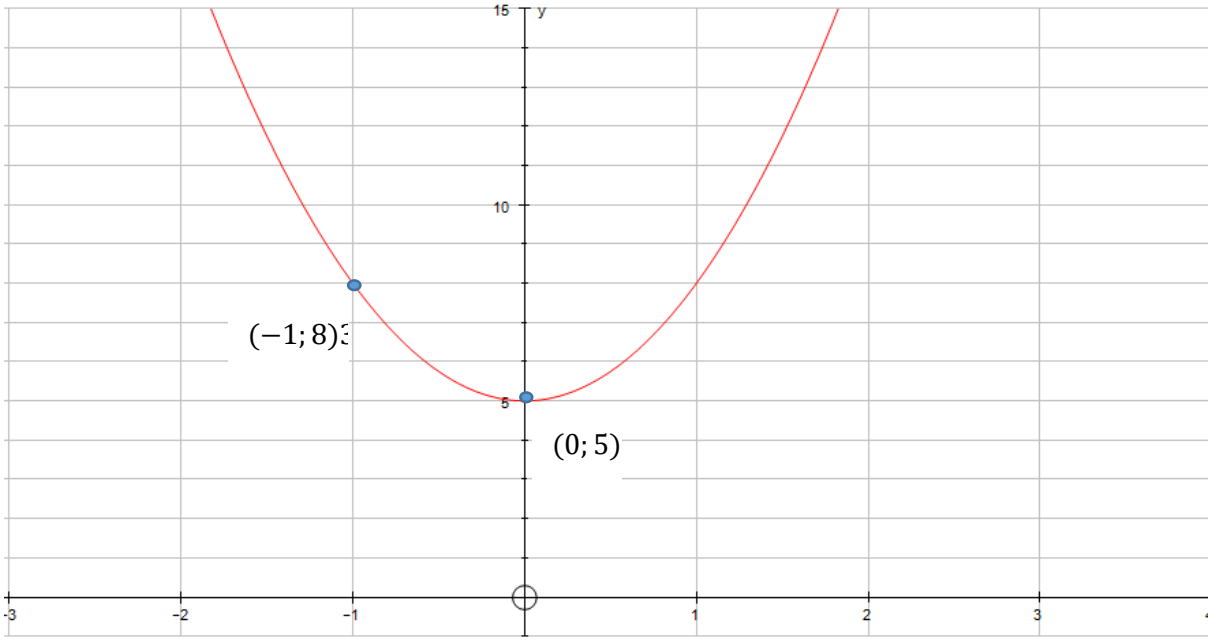
4. GO AHEAD [60 MIN]

4. Determine the equation of the following functions in the format $f(x) = ax^2 + q$:

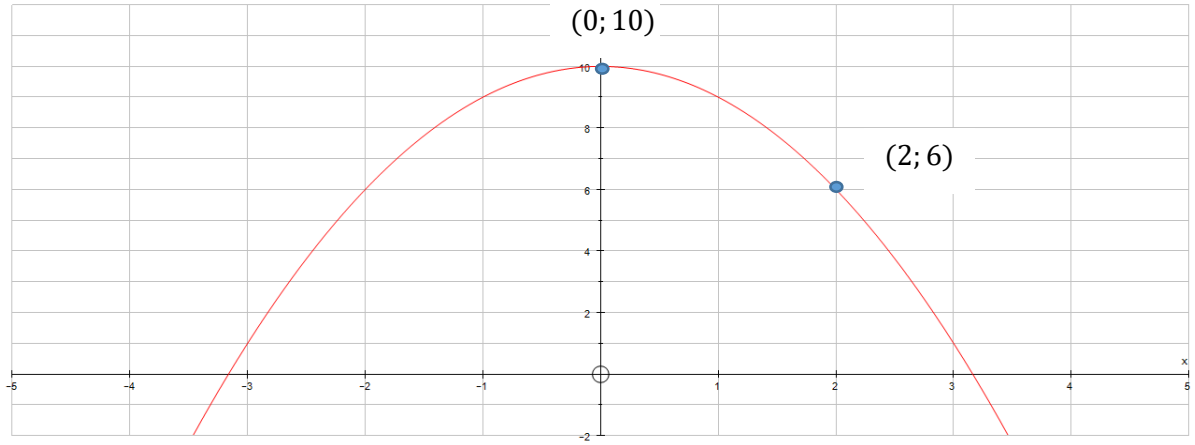
4.1



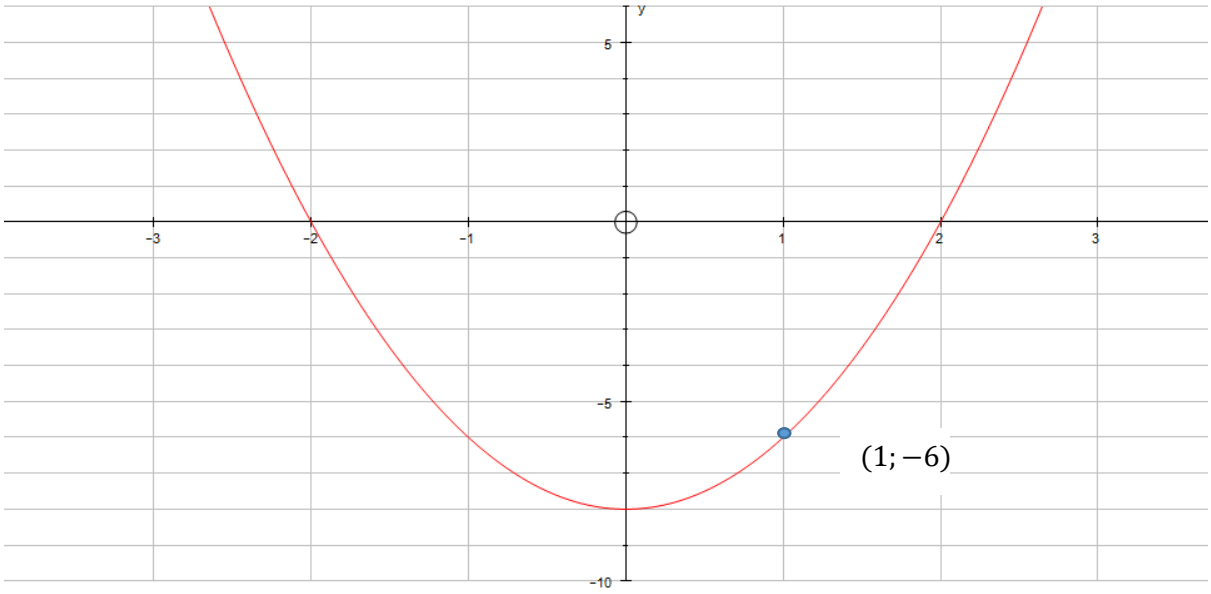
4.2



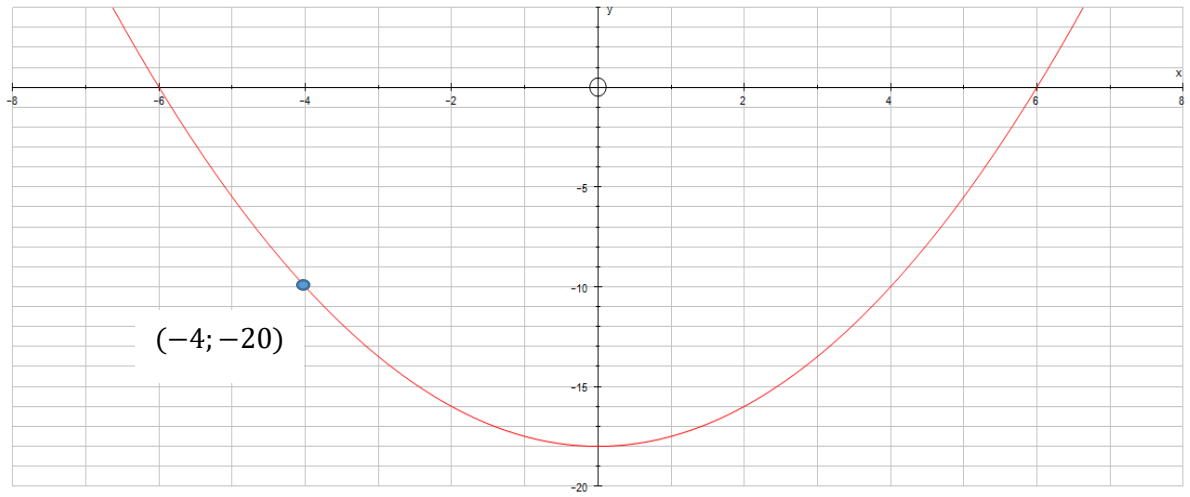
4.3



4.4



4.5



5. CHECK YOUR WORK![15 MIN]

4. HELPFUL VIDEO

https://www.youtube.com/watch?v=jTLuHxxXY_g